

Johann Lee

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Education

Stanford University

Masters, Computer Science

Palo Alto, CA

Incoming Sep. 2026

Cornell University

Bachelor, Computer Science | GPA: 3.97/4.0

Ithaca, NY

Aug 2020 - Dec 2024

• **Honors:** TA of masters' level Machine Learning (CS5780), Former President of Cornell Data Science, Milstein Tech Scholar, 1st @ Teradata Hackathon

Work Experience

Adobe Inc.

Mar. 2026 – Present

Machine Learning Engineer 2, AI Platform for Business

San Jose, CA

- Early promotion to MLE 2 for proactiveness and autonomy. Invited to present projects live to 200+ people at Adobe's AI Day GenAI Showcase.
- Built harness for code generation agent to enable no-code data analysis for 100+ internal users in GTM, saving estimated 3000 hours per quarter.
- Developing computer-use agent to QA product price pages, fixing price discrepancies to reclaim multi-million dollars per year.
- Built agent (similar to legal-AI) to automatically fight payment disputes cases, directly **earning multi-million dollars** per year.

Adobe Inc.

Feb. 2025 – Mar 2026

Machine Learning Engineer, Go-to-Market Growth

San Jose, CA

- Pitched and deployed ML model targeting **hundreds of millions of users** (Acrobat Desktop) for messaging and promos, increasing CTR by 5%.
- Developed ML model to personalize product recommendations at checkout to tens of millions of Acrobat users, increasing mean order size by 4%.
- Proposed and built agentic product-analytics platform to analyze audience segments without coding, raising mean AB test opportunity size by 10%.
- Owned and implemented AI-data-analyst to instantly analyze and interpret AB test results, increasing AB test velocity by 20%.

arXiv.org

Feb. 2024 - Dec. 2024

Research Engineer Intern, Search & Recommendations

Ithaca, NY

- Developed ML-in-the-loop data labeling pipeline to classify research paper submissions reaching 5 million MAU, increasing throughput by 80%.
- Built and trained a real-time recommendation system for paper submission categories, decreasing submission errors flagged by moderators by 10%.
- Fine-tuned LLMs for indexing and retrieval in the document search engine, improving top-5 search hits by 3% over Elasticsearch baseline.

Bank of America

Jun. 2023 – Aug. 2023

Machine Learning Engineer Intern, AI Chatbots

Charlotte, NC

- Built hallucination evaluation and out-of-distribution detection system for [chatbot](#) (40M users), increasing response helpfulness by 20% in internal testing. Business unit estimated \$1M call-center cost savings, work featured at the department's monthly Townhall.
- Researched and implemented hallucination mitigation via paraphrasing, eliminating 40% of hallucination while retaining 90% of correct responses.

Research Publications

[\[ICLR '26\]](#) **Learning From Synthetic Data Improves Multi-hop Reasoning** | *International Conference on Learning Representations*

- We show that RL post-training on synthetic data can induce LLMs to learn generalized reasoning skills in multi-step question answering. Specifically, that it is possible to push LLM reasoning boundaries even by training on incredibly simple data. This enables cheap RL fine-tuning.
- I identified knowledge composition as the key improvement driver and built experiments (& eval suites) showing its persistence even across domains.

[\[ICML '25\]](#) **PhantomWiki: Generating Reasoning and Retrieval Datasets On-Demand** | *International Conference on Machine Learning*

- We built a synthetic-data pipeline for complex LLM reasoning across many documents to address the problem of data-contamination (between public training and evaluation datasets). This enables us to accurately assess the reasoning and retrieval capabilities of popular models.
- I built the document generation pipeline, trained models to show the dataset's contamination-resistance, and built the agentic and RAG systems.

Projects

AI Engineer, Bastion

Aug. 2025 – Feb. 2026

- Built AI platform for commercial real estate brokers across Atlanta, Chicago, and D.C. automating financial modeling and PPT work. [\[Website\]](#)
- Tinkered with prompt prefix caching, language model cascades, task-based model routing, and output caching to reduce LLM API costs by 80%.

Machine Learning Engineer, MathSearch

Jan. 2023 – Jun. 2024

- Led team of 10 to build a cloud-native search engine for math equations, enabling students to search textbooks for complicated formulas. [\[GitHub\]](#)
- Trained vision models to detect equations, fine-tuned image embeddings for similarity search, built the fullstack webapp, and deployed on AWS.

Technical Skills

AI: PyTorch, HuggingFace, MLFlow, vLLM, LangChain, DeepSpeed, FAISS **Languages:** Python, Java, C++, JavaScript, SQL, OCaml, C, CUDA

Frameworks: FastAPI, Flask, React, Django, Spring / Spring Boot, JUnit **Tools:** AWS, Jenkins, Postgres, Docker, Spark, Git, Airflow, MongoDB, Azure